



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT
(BS Computer Science)

COURSE INFORMATION SHEET
(For Lab Based Course)

Session:	Spring-2025
Course Title:	Artificial Intelligence
Course Code:	CS-325L
Credit Hours:	1
Semester:	6 th
Pre-Requisites:	CS-116 Programming Fundamentals
Instructor Name:	Ms. Fozia Khan
Email and Contact Information:	fozia.khan@ssuet.edu.pk ; Room # HS-16, Cubical # 06
WhatsApp Group	Nil
Office Hours:	Monday to Thursday (11:30am to 01:00pm) Or prior appointment
Mode of Teaching:	✓ Synchronous/Asynchronous/ Hybrid/Blended

COURSE OBJECTIVE:

The goal is to acquire knowledge on intelligent systems, what every student should know about Artificial Intelligence. AI is a fast-moving technology with impacts and implications for both our individual lives and society as a whole. In this course, students will get a basic introduction to the building blocks and components of artificial intelligence, learning about concepts like algorithms, machine learning, and neural networks. The course also contains a balanced look at AI's impact on existing jobs, as well as its potential to create new and exciting career fields in the future. Students will leave the course with a solid understanding of what AI is, how it works, areas of caution, and what they can do with the technology.

COURSE OUTLINE:

Introduction to Artificial Intelligence; History and Evolution of Artificial Intelligence, Objectives and Scope of Weak AI and Strong AI; Intelligent Agents, Problem Solving; Problem Representations and its components, Search Techniques; Knowledge and reasoning: Propositional Logic, First-Order Logic, Design & Development of Expert Systems Learning: Learning from Observations, Learning Methods, Supervised and unsupervised Learning, Machine Learning, Neural Networks: Design & Development of Neural Network and its algorithms, Deep learning. Natural Language Processing, Computer vision, Fuzzy Logic: Fuzzy set, operators, classical set.



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT
(BS Computer Science)

COURSE LEARNING OUTCOMES (CLOs):

No.	Course Learning Outcomes (CLOs)	PLOs	Bloom's Taxonomy
1	Follow the instructions to develop working knowledge of Artificial Intelligence, Machine learning, deep learning and natural language processing algorithms in programming languages.	PLO_5 (Modern Tool Usage)	P3 (Guided Response)
2	Construct a project to implement different Artificial Intelligence algorithms in programming language.	PLO_4 (Design/ Development of Solutions)	P4 (Mechanism)
3	Demonstrate the knowledge which will be assessed through viva and presentation.	PLO_7 (Communication)	A3 (Valuing)

RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO-1	CLO-2	CLO-3
Lab Manual	46.43% (13)	16.67% (2)	-
Lab Project	-	83.33% (10)	50% (5)
Final Lab Exam	53.57% (15)	-	50% (5)
TOTAL	56% (28)	24% (12)	20% (10)

GRADING POLICY:

Assessment Tools	Percentage	Marks
Lab Manual	30%	15
Lab Project	30%	15
Final Lab Exam	40%	20
TOTAL	100%	50

Recommended Book:

- Russell, S.J., 2010. Artificial intelligence a modern approach. Pearson Education, Inc..

Reference Books:

- Joshi, P., 2017. Artificial intelligence with python. Packt Publishing Ltd.
- Luger, G.F. and Stubblefield, W.A., 2009. AI algorithms, data structures, and idioms in Prolog, Lisp, and Java. Pearson Addison-Wesley.
- Ross, T.J., 2009. Fuzzy logic with engineering applications. John Wiley & Sons.



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT
(BS Computer Science)

BENCHMARKS:

- **University of Sargodha:**

<https://www.studocu.com/row/document/comsats-university-islamabad/advanced-topics-in-ai-and-neural-computing/1586810323-lab-manual-for-all-weeks/19227181>

- **BRCM College of Engineering & Technology:**

<https://www.brcmcet.edu.in/downloads/files/n64007aaa14a24.pdf>

- **Dronacharya College of Engineering:**

<https://ggnindia.dronacharya.info/IT/Downloads/Labmanuals/AI-VI-11072023.pdf>

- **Botswana International University of Science & Technology:**

<http://repository.biust.ac.bw/handle/123456789/414?show=full>

JOB DEMAND:

- AI/Machine Learning Engineer
- Data Scientist
- NLP Engineer
- Data Analyst
- Big Data Engineer
- AI Software Engineers



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT
(BS Computer Science)

COURSE BREAKDOWN WITH LAB SYNCHRONIZATION:

- **Both sides same Colours:** Lab is synchronized with the topic
- **Red Color:** Lab is not synchronized (*conducted before theory*)
- **No Color:** Lab is to introduce new hardware or software skill /
Open Ended Lab / Lab is relevant to a topic taught in
pre-requisite and required for upcoming labs.

Week No.	Topics	Laboratory Synchronization
1	Module I: Introduction to Artificial Intelligence. The Foundations of Artificial Intelligence / Areas and Application of AI, History & Evolution of AI.	Familiarization with Python language and how to use different syntax to control indexing using strings, loop and conditional statement
2	Module II: Perception and Intelligence. Agents Architecture and types, Good Behavior: The Concept of Rationality	Familiarization with Python language using functions, list, tuple and dictionary
3	The Structure of Agents, Agents Algorithms, PEAS, Generic Agent, Autonomous Agent, Types of Agent programs	Familiarization with the concepts of class and object using Python language
4	Solving Problems by Searching, Problem-Solving Agents, Example Problems	Manage a set of records with priority queue using queue and heapq module in python
5	Module III: Searching Algorithms in AI. Uninformed Search Strategies, BFS, DFS, Uniform cost search.	Implement shortest path using Breadth First Search (BFS) and Depth First Search (DFS) techniques in python
6	Informed (Heuristic) Search Strategies Greedy Best First Search, A* search.	Implement informed A* Search, Best First Search and Greedy Best First Search algorithms using Python
7	Local Search Algorithms and Optimization Problems, Local search in continuous space, hill climbing and genetic algorithm.	Familiarize students with the NumPy and Matplotlib libraries of python
8	Adversarial Search, Game theory, optimal decisions in games, minimax, alpha-beta tree search.	Familiarize students with prolog programming, rules and facts. Also exploring operators and built in predicates of prolog
9	Mid Term Examination	
10	Module IV: Knowledge representation. Introduction to Logic, Propositional Logic: A Very Simple Logic.	Perform supervised learning Decision Tree and Naïve Bayes algorithms for training, testing and classification using Python



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT
(BS Computer Science)

11	First-Order Logic, Syntax and Semantics of First-Order Logic, Using First-Order Logic, Overview of Expert System Technology, Rule Based Expert Systems.	Perform unsupervised learning K-means clustering algorithm for training, testing and classification using Python
12	Module V: Machine Learning. Learning & its Method, Supervised, unsupervised and reinforcement learning, Linear Regression and Classification.	Implement Artificial Neural Network (ANN) and Convolutional Neural Network (CNN) using Python
13	Module VI: Deep Learning & Neural Networks. Artificial Neural Networks, Simple Feedforward Networks, Deep Learning, Back propagation, Generalization.	Implement various functions related to Natural Language Processing (NLP) using Python
14	Convolution Neural Networks Unsupervised Learning and Transfer Learning, Applications.	Getting started with OpenCV on Jupiter Lab
15	Module VII: Natural Language Processing. Grammar, Parsing Augmented Grammars, Complications of real Natural Language. Naïve bayes for natural language processing.	Open Ended Lab
16	Module VIII: Fuzzy Logic. Introduction to Fuzzy Logic, Difference between Crisp set and fuzzy set. The Utility and limitations of Fuzzy Systems, Fuzzy Sets.	Project Demonstration
17	Revision	Final Lab Exam and Viva



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT
(BS Computer Science)

LAB PLAN

Course Title: Artificial Intelligence

Course Code: CS-325L

Week No.	Lab Date	Objective	Required Reading
1	03-03-2025 to 07-03-2025	Familiarization with Python language and how to use different syntax to control indexing using strings, loop and conditional statement	Lab Manual
2	10-03-2025 to 14-03-2025	Familiarization with Python language using functions, list, tuple and dictionary	Lab Manual
3	17-03-2025 to 21-03-2025	Familiarization with the concepts of class and object using Python language	Lab Manual
4	24-03-2025 to 28-03-2025	Manage a set of records with priority queue using queue and heapq module in python	Lab Manual
5	31-03-2025 to 04-04-2025	Implement shortest path using Breadth First Search (BFS) and Depth First Search (DFS) techniques in python	Lab Manual
6	07-04-2025 to 11-04-2025	Implement informed A* Search, Best First Search and Greedy Best First Search algorithms using Python	Lab Manual
7	14-04-2025 to 18-04-2025	Familiarize students with the NumPy and Matplotlib libraries of python	Lab Manual
8	21-04-2025 to 25-04-2025	Familiarize students with prolog programming, rules and facts. Also exploring operators and built in predicates of prolog	Lab Manual
9	Mid Term Examination (28-04-2025 to 02-05-2025)		
10	05-05-2025 to 09-05-2025	Perform supervised learning Decision Tree and Naïve Bayes algorithms for training, testing and classification using Python	Lab Manual
11	12-05-2025 to 16-05-2025	Implement unsupervised learning algorithm (K-means clustering) for training, testing and classification.	Lab Manual
12	19-05-2025 to 23-05-2025	Implement Artificial Neural Network and Convolutional Neural Network and analyze the performance measures of ANN and CNN.	Lab Manual



(SSUET/QR/111)

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT
(BS Computer Science)

13	26-05-2025 to 30-05-2025	Implement Natural Language Processing using NLTK library.	Lab Manual
14	02-06-2025 to 06-06-2025	Getting started with OpenCV on Jupiter Lab	-
15	09-06-2025 to 13-06-2025	Open Ended Lab	-
16	16-06-2025 to 20-06-2025	Project Demonstration and Viva	-
Lab Examination and Viva (23-06-2025 to 27-06-2025)			